Abstract

Computer networks and systems are seriously threatened by DoS (Denial of Service) attacks, especially SYN flood attacks. By flooding the target system with SYN packets and taking advantage of a flaw in the TCP handshake procedure, SYN flood attacks exhaust the target system's resources and make it unreachable to authorised users. To investigate and evaluate the SYN flood attack and how it affected network performance. The project will entail running tests that mimic SYN flood attacks on a target system, gathering and analysing network traffic data, and assessing the efficiency of various detection and mitigation strategies.

Simulating the SYN flood attack by using Hping3 tool which is pre-installed in Kali Linux. To protect other systems from any potential threat, this system will be isolated from the rest of the network. Windows will be the target machine Kali Linux will be the attacking machine. Both are connected to the same subnet, thus it will be able to analyse traffic using Wireshark. A port scan will be required to discover an open port on the target machine, Nmap is used for port scanning.

The findings indicated that the SYN flood attack significantly impacted the target system, leading to longer response times, higher error rates, and clogged networks. For reducing the effects of SYN flood attacks, countermeasures including firewalls, load balancers, and restricting concurrent connections were shown to be successful. To improve defences against these cyberthreats, this research offers insightful information about the dangers and mitigation tactics related to SYN flood attacks.

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List of Abbreviations

DOS Attack: - Denial of Service Attack

SYN Attack: - Synchronise Attack

TCP: - Transfer Control Protocol

UDP: - User Datagram Protocol

ICM: - Internet Control Message Protocol

DDOS: - Distributed Denial of Service

IDPS: - Intrusion Detection and Prevention System

SIEM: - Security Information and Event Management

ACL: - Access Control List

ACK: - Acknowledgement

RTT: - Round Trip Time

HAProxy:- High Availability Proxy

NLB: - Network Load Balancing

OPNET: - Optimized Network Engineering Tools

ANN: - Artificial Neural Network

PPTP: - Point-to-Point Tunneling Protocol

VPN: - Virtual Private Network

SDN: - Software-Defined Networking

Nmap: - Network Mapper

1. INTRODUCTION

One of the most common security intrusion techniques used by attackers is the Denial-of-Service attack, which aims to bar authorised users from using the targeted host or other network resources. In order to cause the victim's host to run out of resources, a denial-of-service attack typically involves flooding the server or host (memory, CPU, traffic). It cannot accommodate legitimate users due to this problem. The result of a successful flooding attack is that the victim's host becomes unable to accommodate legitimate users, as the resources are consumed by the flood of malicious traffic. Unlike other security threats that can be mitigated by simply rebooting the affected system, flooding attacks are often challenging to defeat and require specialized mitigation techniques (Sumantra & Indira Gandhi, 2020). Flooding attacks can take different forms, such as SYN flood attacks, UDP flood attacks, and ICMP flood attacks, among others. These attacks can cause severe disruptions to online services, websites, or other network resources, resulting in downtime, loss of revenue, and damage to the reputation of the targeted entity. Flooding is challenging to defeat and cannot be fully mitigated by rebooting, unlike other threats. Mitigating flooding attacks requires proactive measures such as rate limiting, traffic filtering, and implementing SYN proxy, among others. These measures aim to detect and block malicious traffic, limit the rate of incoming requests, and validate and filter out illegitimate packets before they reach the targeted host or network resources. (About Flood Attacks, n.d.)

A SYN flood, often referred to as a TCP SYN flood, is a sort of distributed denial-of-service (DDoS) or denial-of-service (DoS) attack that sends a server a large number of SYN requests to overload it with open connections. SYN floods, also known as half-open attacks, involve flooding a server with connection requests while ignoring the corresponding acknowledgements. Since there are so many open TCP connections as a result, the server's resources are quickly depleted, effectively crowding out legitimate traffic, making it impossible to establish new, authorised connections and making it difficult or impossible for those who are already connected to the server to use it properly. This kind of attack could potentially affect any company having a website that is accessible to the public(What Is a TCP SYN Flood DDOS Attack? | Radware, n.d.) .

A SYN flood can quickly overwhelm a server, severely impede server answers, and block any additional connections if it is not quickly noticed and handled. This basically shuts down the server, denying service to genuine users, restricting access to apps and data, or stopping online shopping. A loss of business continuity, an interruption of vital infrastructure, lost revenue, or

a tarnished brand are possible outcomes. The consequences of missing data access can be fatal for some businesses, such those in the healthcare sector (*What Is a SYN Flood Attack?*, n.d.).

A SYN flood attack can significantly affect a target system's and network's performance or possibly cause it to shut down entirely. Monitoring the quantity of successful connections, unsuccessful connections, and response times will allow to gauge the impact of the attack (POPOVICI, 2023). A SYN flood attack often has a relatively small percentage of successful connections while having a large percentage of unsuccessful connections. As it takes longer to handle the influx of connection requests, the system can also experience slower response times. Furthermore, the assault may use a sizable amount of bandwidth, thus impairing network performance (Eddy, 2007).

As a result, a complex environment for cyber security has emerged as a result of technological improvement and the dissemination of more criminal attacks. Thus, the proposed effort considers the challenges raised by focusing on the following research concerns:

- A review of the DoS attack and its application using the hping3 tool. This makes it crystal clear how a DoS assault is carried out by an attacker.
- The variation is described in the results when the attacks are posed against the three restrictions of CPU usage, frame duration, and flow degree.

E-learning companies, e-marketing businesses, and e-health care providers are just a few examples of businesses based on internet services that have been expanding steadily and exponentially. Contrarily, there has been a significant rise in usage of Facebook, Twitter, WhatsApp, and other social networking sites. Cloud-based service providers, like Google, Amazon, and Azure, are currently multiplying exponentially. Due to the enormous traffic that is produced by the Internet's dramatic growth in users, public network abuse has increased. In light of this, it is possible that online services will occasionally be unavailable. Even some highly protected, sophisticated systems are vulnerable to attack (Zebari et al., 2020). It will be easier to defend the attacked system if you are aware of the attack and how it works. It helps the organisations, employees or users who are legitimate users to understand how denial of service attack perform over a network.

Common DOS Attacks

a. Slow Loris Attack

A webserver-marking attack method that enables a single machine to take out the target web server without affecting other services or ports. Such attacks make use of sparse bandwidth. It has been discovered that slow loris attacks are very prevalent against Apache 1.x and 2.x. Researchers have likened a sluggish loris attack to a turtle's victory in a race despite being slow and stable. The aim behind this low and slow attack is to keep the connections open for a lengthy amount of time, and this may be accomplished by sending fractional requests. Slow loris does the same thing by sending the request as slowly as possible (What Is Slowloris DDoS Attack Mitigation Methods | Wallarm, n.d.).

b. Ping Flood Attack

An easy-to-use but powerful strategy for attack. The ICMP or ping packets, which are typically used in ping flood attacks, are sent in excess to the target by the hacker. An attacker can broadcast many ping packets at once by using the ping command. the attacker sends ping packets to the target host without waiting for responses or acknowledgments from the other side. This is done to maximize the volume of attack traffic and to consume the target's resources, such as network bandwidth, processing power, and memory, as quickly as possible. The attacker may broadcast or multicast the ping packets, or send them to multiple hosts simultaneously, amplifying the impact of the attack. Not waiting for responses or a response from the other side is the preferred strategy. The ping flood attack is considered a powerful strategy due to its simplicity and ease of use. It does not require sophisticated techniques or tools and can be executed using basic command-line utilities. The attack can quickly overwhelm a target host, making it unresponsive or unavailable to legitimate users, resulting in service disruption or denial of service (Stiawan et al., 2021).

c. SYN Flood Attack

Making the service unavailable to its actual traffic is the main goal of this attack. Although the SYN flood attack doesn't believe in establishing a connection, this may be accomplished by tricking the TCP handshake procedure. For the purpose of connecting the servers, it initially sends SYN packets. A packet known as a SYN/ACK packet is sent back in response by the server to that initial request in order to confirm the connection that is just beginning. Following the phase of client confirmation, the connection has already been validated. This is

accomplished by sending an ACK packet along with the acknowledged ACK message (Al-Duwairi et al., 2020).

Mitigation Stages

The term "mitigation" refers to a straightforward reduction in the unfavourable effects of a condition. A DDoS attack may be defended against in four stages. According to Figure 1, there are four steps to reducing a DDoS attack.

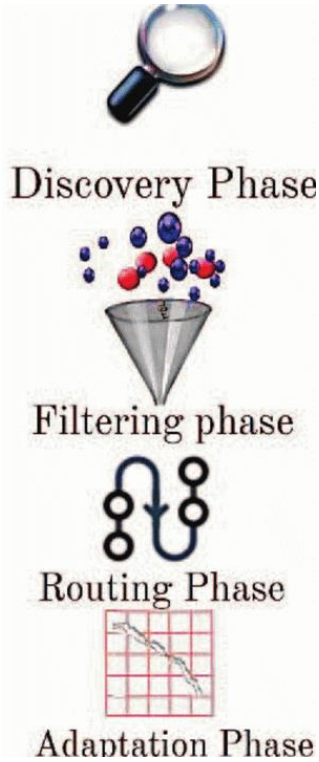

Figure 1: Mitigation Phases (Kukreti et al., 2022)

Discovery - The first step is a discovery phase, where the main goal is to identify the malicious traffic and the traffic from potential customers in order to detect the attack. By distinguishing between a crowd with a high volume and one with a normal volume, this may be examined. Network traffic is monitored using various techniques, such as intrusion detection and prevention systems (IDPS), security information and event management (SIEM) systems, or network monitoring tools. It is important to have robust traffic monitoring and analysis mechanisms in place, along with well-defined incident response procedures, to effectively detect and respond to potential attacks and protect the system and data (What Is DDoS Mitigation? Complete Guide (2023), 2022).

Filtering - The second phase of mitigating a potential attack after the discovery phase is to respond to the genuine population by filtering out all harmful traffic. In order to minimise bottlenecks, the second phase responds to the genuine population by screening all harmful traffic. Filtering is an important step since the goal is to make the service available only to the intended audience. Traffic is filtered to separate legitimate traffic from malicious traffic. This can be done using various techniques, such as firewall rules, access control lists (ACLs), or security policies, to block or allow traffic based on specific criteria, such as IP addresses, protocols, ports, or other parameters.

Routing – Routing plays a crucial role in mitigating the impact of heavy traffic loads caused by attacks, as it can help break up the heavy load into manageable chunks and reroute traffic to different paths. This can help prevent bottlenecks and reduce the impact of the attack on the targeted systems or services. Routing can assist in breaking up the heavy load into manageable chunks. Rerouting requires a decision to be made on whether to filter or reject the crowd. Traffic shaping is a technique that prioritizes or throttles certain types of traffic to control the flow of data and prevent congestion. By shaping the traffic, organizations can control the rate at which traffic is sent or received and prioritize legitimate traffic over malicious traffic. This can help in managing the heavy load caused by an attack and prevent the targeted systems or services from being overwhelmed.

Adaptation- The adaptation phase in the context of mitigating a syn flood attack involves analysing the typical attack pattern used by an attacker and making adjustments to the defense mechanisms based on this analysis. Analysing the typical attack pattern used by an attacker is the goal of the adaptation phase. By limiting traffic originating from a certain area or restricting an abusive IP address, it is possible to identify this recurrent trend based on repeated statistics. Identifying recurrent trends in the attack pattern by analyzing repeated statistics. This may include analyzing the source of the attack traffic, such as the geographical location or IP address from which the attack is originating. By identifying common patterns in the attack traffic, such as a high volume of traffic coming from a specific geographic area or a certain IP address, it makes adjustments to their defense mechanisms to specifically target and block this traffic. (DDoS Mitigation | How to Choose the Right Mitigation Service | Imperva, n.d.).

1.1Background of the Project

The project centers around the increasing threat of SYN flood attacks, which are a type of denial of service (DoS) attack that targets the TCP handshake process. In a SYN flood attack, an attacker floods a target system with a massive number of SYN packets, overwhelming the system's resources and causing it to become unresponsive or unavailable to legitimate users.

Three-Way Handshake scenario

A three-way handshake is used to start TCP traffic. By requesting a communication session with the server during the TCP handshake phase shown in Figure 2 the client must start the conversation:

1. The client sends a segment with SYN and notifies the server that the client should begin communication and what its sequence number should be.
2. SYN-ACK signal is set when the server answers to a client request. SYN indicates what sequence number it should be able to start with the segments, and ACK helps to denote the response of the segment that is received.
3. Finally, the client returns an ACK packet.

The TCP connection is then active and ready to send and receive data (Williams, 2022).

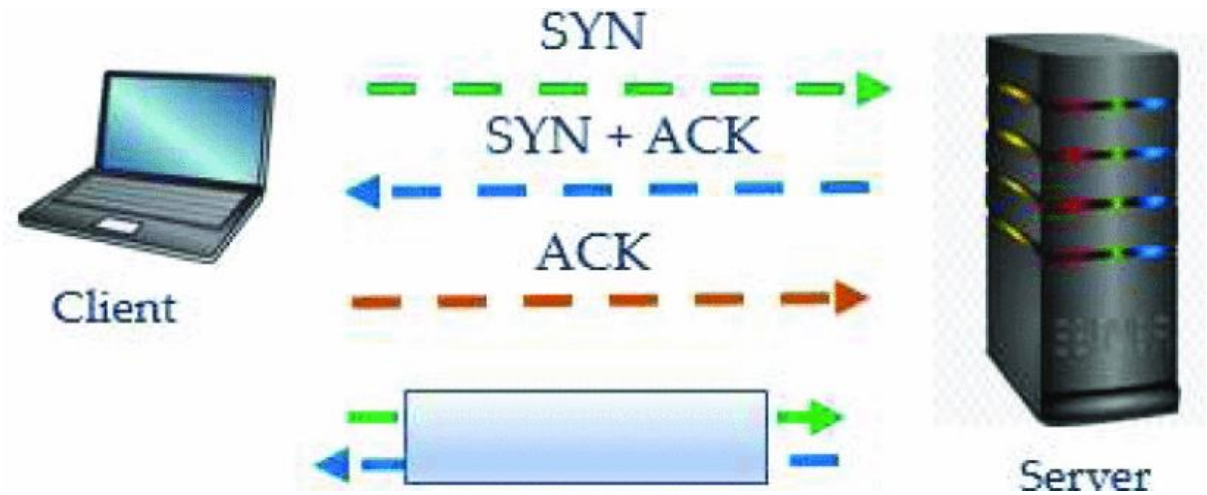

Figure 2: Normal TCP 3-Way Handshake (W Mariam & Negash, 2021)

SYN Flood Attack Scenario

A stateful protocol attack known as a "SYN flood" occurs when a requester repeatedly sends SYN requests to a server without first sending SYN/ACK requests and waiting for a response from the server (Villing, 2019).

1. A large number of SYN packets with spoof IP addresses are sent by the attacker to the victim server.
2. A SYN/ACK packet is then returned by the victim server.
3. The final ACK from the attacker never replays to the victim server; as a result, a lot of half-open connections are kept in the victim server's queue, and it fills up quickly.

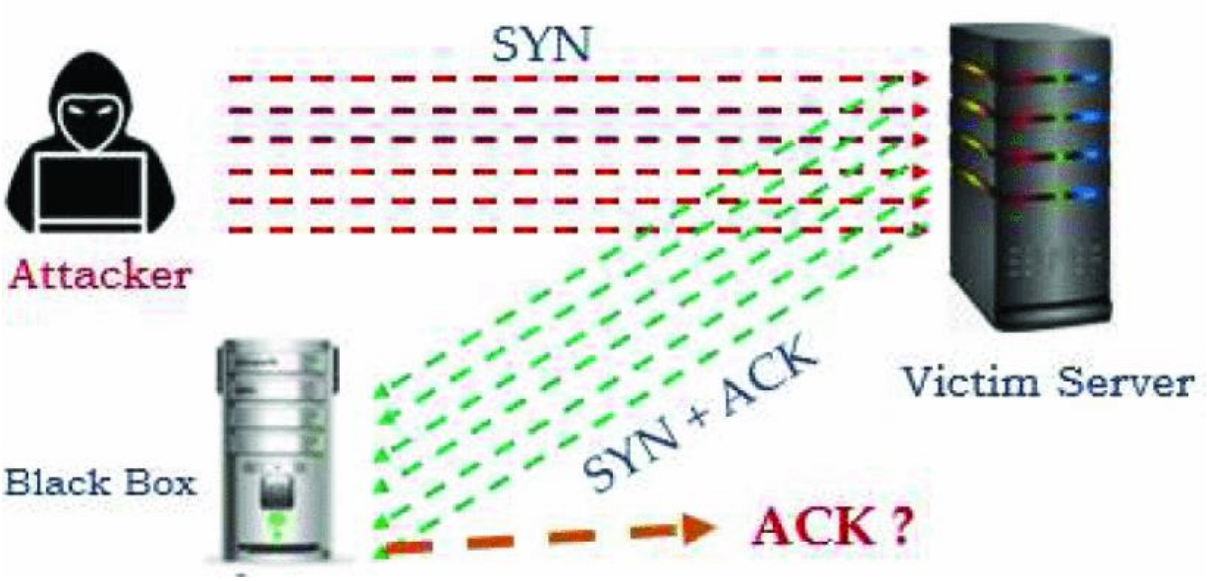

Figure 3: SYN Flood Attack Scenario (W Mariam & Negash, 2021)

Characteristics of SYN Flood Attack

These are some of the salient features of the SYN flood attack:

- Distributes a large number of packets in a short time frame.
- It has smaller, fixed-size packets.
- The time between packets is almost zero.
- It has exclusively utilised TCP protocols.
- Many half-opened connections (W Mariam & Negash, 2021).

1.2 Project Objectives

A comprehensive understanding of the SYN flood attack, including how it works, its impact on target systems, and the need for effective mitigation measures. Measure and analyze performance metrics such as Round-Trip Time (RTT), CPU utilization, and network bandwidth before, during, and after SYN flood attacks to evaluate the impact of the attacks and the

effectiveness of the mitigation measures. Set up rate limiting rules to restrict the number of connections allowed from a single source IP address within a certain time frame, and evaluate its effectiveness in mitigating SYN flood attacks. Analyze the data collected during the experiments and interpret the results to draw conclusions about the effectiveness of rate limiting and SYN proxy as mitigation measures against SYN flood attacks. The findings including the effectiveness of rate limiting and SYN proxy as mitigation measures, and provide recommendations for securing systems against SYN flood attacks based on the research findings.

- To comprehend the fundamentals of SYN flood attacks, including their features, classifications, and typical TCP/IP protocol flaws.
- To build up a lab environment that mimics a network architecture and permits the deployment of SYN flood attacks.
- To perform a SYN flood attack on a target system in a lab setting and assess the effect of the attack on the target system and the network as a whole.
- To examine the defences against SYN flood attacks and mitigations that may be applied, such as firewalls, intrusion detection systems (IDS), and load balancers, in order to avoid or reduce the effects of these attacks.
- The implications of the findings for network security and the bigger picture of cybersecurity, including possible dangers and threats posed by SYN flood attacks and the significance of adopting proactive efforts to stop and mitigate such attacks.

2. LITERATURE REVIEW

Denial of Service Attack

A technique for reducing and preventing HTTP DOS assaults. They used HAProxy as a defence method against this kind of assault. They used the goldeneye tool to generate bogus HTTP requests, and Apache was their web server of choice. The experiment demonstrated that the web servers were unable to offer any service after launching an attack without HAProxy for 5 seconds. Although the HAProxy restricted the majority of the attacker's connections, it also stopped HTTP attacks from using backend web server resources. The Logstash, Elasticsearch, and Kibana components of the elastic stack architecture were used to gather logs from the web server, filter and query the logs, and then show findings in dashboards and graphs to aid in the detection of an attack through visual log data analysis. To aid in the detection of attacks, the

researchers employed the Logstash, Elasticsearch, and Kibana components of the elastic stack architecture to gather logs from the web server, filter and query the logs, and visualize findings in dashboards and graphs. This allowed for visual log data analysis to identify any abnormal patterns indicative of an attack. The researchers also established guidelines to notify the company when abnormalities were detected based on pre-established benchmarks. Guidelines are also set up to notify the company when abnormalities are found based on pre-established benchmarks (Ezenwe et al., 2020).

The authors investigate the vulnerabilities and risks posed by DoS attacks on power systems, analyse the potential consequences, and propose mitigation strategies to enhance the cyber security of power systems. First, the authors develop a state model for a closed-loop power system in one area, taking into consideration the time delay of communication channels. Secondly, they integrate an event-triggering control mechanism with the load frequency control scheme of the power system, aimed at improving the utilization of communication channels for transmitting area control error. They utilize the average dwell time design approach to obtain exponential stability criteria and ensure good stability effect for the one area power system with an event-triggered load frequency control scheme under denial-of-service attacks. This can be achieved by choosing an appropriate rate for the allowable duration of denial-of-service attacks during system operation and obtaining a time delay margin for these stability criteria (Shen et al., 2019).

In this paper, the author presents an attack Models and Security Analyses" likely provides an overview of denial-of-service (DoS) attacks specifically targeted at control systems, which are systems used to monitor and manage critical infrastructure such as power grids, transportation systems, and industrial processes. A particular focus is placed on tail-probability based failure models, which have recently been used to describe jamming attacks that affect wireless channels' signal to interference-plus-noise ratios as well as transmission failures on multi-hop networks brought on by packet-dropping attacks and benign problems. Attack models are used in networked system security assessments, providing insights into the current advancements in the design of attack-resistant control and communication protocols. The author likely discusses various modeling and analytical issues related to DoS attacks in control systems, including the challenges and considerations in protecting critical infrastructure against such attacks. The paper likely provides a comprehensive review of the latest research and developments in the field, highlighting the efforts made in designing robust and resilient control and communication protocols to mitigate the impact of DoS attacks on control systems. Provide an explanation of

how attack models are used in networked system security assessments. A review of the most current advancements in the design of attack-resistant control and communication protocols is included in addition to the modelling and analytical issues (Cetinkaya et al., 2019).

SYN Flood Attack

The authors investigate the effectiveness of various sniffing tools, such as Wireshark, TCP dump, and Snort, for detecting different types of DoS attacks, and analyze their advantages and limitations in different network scenarios. The problem of DoS attacks, which are a common form of cyber-attack that aim to disrupt the availability of a network or system by overwhelming it with a flood of malicious traffic. The authors highlight the importance of timely and accurate detection of DoS attacks to mitigate their impacts and protect network resources and services. Data gathering, identification, and an analysis of an attacker's characteristics are the two primary aspects of the approach. With two nodes, one plays the role of an attacker machine, and the other plays the victim computer with a programme installed to capture all network traffic entering the network environment, the behaviour of assaults are identified. Users are unable to access internet services because networks become less responsive when odd harmful activity occurs. The methodology for evaluating the effectiveness of the sniffing tools in detecting different types of DoS attacks. Describe the scenarios that used for the experiments and the metrics employed to measure the performance of the tools, such as detection accuracy, false positives, and false negatives (Alexis Fidele et al., 2020).

This paper focuses on the impact of SYN flood Distributed Denial of Service (DDoS) attacks on web servers that are configured in a HAProxy and NLB cluster-based architecture. It is effectively examined how Internet Information Service 10.0 (IIS 10.0) on Windows Server 2016 and Apache 2 on a Linux server running Ubuntu 16.04 perform. Both network load balancing (NLB) and high available proxy (HAProxy), which are used for web server load balancing in the Windows and Linux systems, are subject to performance testing. Also, the research assessment measures affect the web servers' dependability, effectiveness, and responsiveness. The article also measures the throughput of both techniques and the average CPU consumption. The findings demonstrate that the IIS 10.0 cluster-based web servers are more responsive, effective, and stable against SYN flood DDoS attacks than they are under other types of attacks. In terms of average CPU utilisation and throughput, the IIS 10.0 web server performs better than Apache 2 (Zeebaree et al., 2020).

In this article the author illustrates various defence mechanisms, including TCP intercept, rate limiting, ACL and Reverse Path Forwarding, and evaluate their effectiveness in mitigating TCP SYN flood DDoS attacks. The greatest protection was determined to be router-based TCP Intercept, whereas the weakest defensive was Anti DDoS Guardian. It was determined that the SYN attack's most damaging aspect was its consumption of network capacity, which caused genuine users to wait longer to reach the target system. TCP Intercept, which represents the server, is the best choice for defence against network consumption and appliances. Despite this, the average delay grew from 1.92 ms (without attack) to 3.21 ms (with defence). TCP intercept decreased the victim's CPU use to 1%, while the traffic bandwidth rate only slightly rose from 1.342 kbps (before the assault) to 1.462 kbps (with defence) (Kolahi et al., 2014).

The study examines the systems susceptibility to TCP (Transmission Control Protocol) segment targeting while the SYN flag is ON, which creates a window for a DoS (Denial of Service) attack known as SYN flooding assault, or more often known as a SYN flood attack. In a simulated environment called OPNET, the impacts of this kind of attack are examined and displayed. The experiments in this simulated environment to evaluate the effects of SYN flood attacks on the performance and availability of the targeted systems. This may include measuring metrics such as network throughput, response time, and resource utilization to assess the severity of the attack and its impact on the system's performance. To defend against SYN flood attacks, the study likely proposes and evaluates two anomaly detection techniques. These techniques may involve analyzing network traffic patterns and behaviors to detect abnormal patterns that may indicate a SYN flood attack in progress. The researchers may compare the performance and effectiveness of these techniques in detecting and mitigating SYN flood attacks and provide insights into their strengths and limitations. As a strong defence against this kind of attack, two anomaly detection techniques are used. Also, useful SYN flood attack defence strategies for Linux and Windows environments (Bogdanoski et al., 2013).

SYN Flood Attack using Metasploit

The article describes several DDoS attacks, their mitigation steps, how attacks spread, malicious codes, and then offers solutions for displaying both typical and DDoS attack situations. It has been demonstrated how to use Metasploit to exploit a SYN flooding attack. In both the normal scenario and attack scenario, the CPU's usage of frame length and rate has been noted. In a typical situation, CPU utilisation is roughly 20%, according to preliminary findings. The CPU execution overhead in attacked phases, on the other hand, is close to 90%

or 100% with the same CPU load. As a result, the investigation revealed that the CPU utilisation, frame duration, and level of data flow were significantly different. A network traffic analyser tool has been utilised, Wireshark (Kukreti et al., 2022).

The paper presents a real-world case study of a SYN flood attack that was launched using Metasploit, which is a popular penetration testing framework widely used for ethical hacking and security testing purposes. In this work, the victim system runs Windows 2008, and the Kali Linux computer is utilised as a virtual machine to simulate an attacker attacking the victim machine. The well-known operating system Kali Linux is used by both unethical and ethical hackers to conduct pen-testing and hack victims. And there is a tool that is used in Kali Linux to carry out the SYN flood attack. This tool is the Metasploit framework, which is also a well-known penetration testing framework that is currently used by all professional ethical hackers as well as the unethical hackers because it is simpler to use and has a command-line interface that is more professional than the tools that have a graphical user interface. If hackers are aware of the IP address of the target computer, they may quickly begin a SYN flood assault and send all SYN packets to the victim system in an effort to jam it (Zuin et al., 2021).

SYN Attack using Hping3

The authors conduct an empirical study to assess the effectiveness and performance of different machine learning techniques in detecting SYN flood attacks in network environments. the evaluation of categorization machine learning (ML) algorithms' performance in detecting SYN flood attacks. The Hping3 and Wireshark programmes are used to generate and collect packets from the ethio telecom network, respectively, and the resulting dataset is used to train and test the classification models. Four classification ML algorithms and three training methods have been used to further pre-process and analyse this dataset. The WAKA (Waikato Environment for Knowledge Analysis) data mining tool has been used to carry out the implementation. AdaBoost, Naive Bayes, and ANN algorithms perform with 98.52%, 95.31%, and 94.85% accuracy, respectively, while the J48 method performs with 98.57% accuracy. Consequently, based on the performance a model using the J48 algorithm has been proposed for SYN attack detection (W Mariam & Negash, 2021).

The author focuses on investigating the impact of a Distributed Denial of Service (DDoS) attack on PPTP VPN, which is a type of personal point to point virtual private network protocol commonly used for remote access and secure communication over the internet. In order to enable access for Windows 10 clients, the author suggested that a Direct Access PPTP VPN be

set up and configured with a default VPN gateway on a Windows Server 2016 system running Routing, Remote Access, and Active Directory. The client is then set up to connect remotely across a network to this VPN. Four Kali Linux workstations are using a Hping3 Denial-of-Service attack against the client on the same network to bombard it with packets and stop it from providing the service. As a result, it can be seen how the VPN responds to the assault and what steps it takes to shield the user from future ones. The VPN failed to connect when the client was being attacked, it was discovered. In addition, if the client is attacked while already connected to the VPN, the VPN disconnects the client. This connects to the infamous vulnerability of PPTP and explains why alternative protocols have been more extensively adopted. It is also accompanied by nearly total network instability and packet loss (Jones et al., 2019).

TCP SYN flooding is regarded as one of the most efficient attacks in software-defined networking (SDN) to conduct control plane and target server saturation. A significant number of malicious SYN requests are generated by the attacker in this attack, and since there are no forwarding rules, the data plane switches must transmit these SYN signals to the controller. The communication link between the data plane and the control plane becomes congested as a result of this excessive forwarding, and both planes' computing capacity is depleted. To identify and stop SYN flooding in SDN networks, we provide an unique countermeasure in this study dubbed SYN-Guard. Implementing SYN-Guard into practise on the SDN controller to examine incoming TCP connection requests. The controller sets up forwarding rules for SYN requests that successfully pass SYN-validation Guard's test. After identifying the source of the bogus SYN request, SYN-Guard stops it from making any more SYN requests to the data plane switch on that host. When compared to conventional SDN and state-of-the-art proposals, SYN-Guard reduces the average response time during an ongoing SYN flooding attack by up to 21%, according to the performance evaluation carried out using the simulation results. SYN-Guard also exhibits low side effects for genuine TCP requests (Mohammadi et al., 2019).

SYN Flood Attack Mitigation

The authors likely propose a detection and mitigation technique that involves leveraging statistical analysis techniques to identify patterns and anomalies in network traffic associated with SYN flood DDoS attacks. The paper provides an overview of the SDN environment, which is a network architecture that allows for centralized control and management of network resources using programmable controllers. SDN makes use of the concepts of centralised

management and data plane isolation from the control plane. The centralised controller makes it simple to set policies, while SDN is more flexible. Due to SDN's improved security, it is now frequently employed in contemporary network paradigms. To recognise and counteract TCP SYN flood DDoS assaults, this article proposes an entropy-based statistical technique. In order to reduce the number of false positives, the suggested method employs a three-phased detection strategy. It has proven possible to identify intrusions using entropy, standard deviation, and weighted moving average. The results of several tests indicate that the recommended strategy is less risky, more portable, and less likely to produce false-positive findings (Batool et al., 2022).

A detection mechanism that involves monitoring the self-information, which is a measure of the randomness or unpredictability of the network traffic, at different hierarchical levels in the network. The self-information may be calculated based on statistical properties of the traffic, such as entropy or information theory measures. The suggested methodology, which offers low misjudgement, a hierarchical detection strategy, and minimal time complexity, is more effective at detecting the SYN flooding attack than existing entropy measurement detection approaches. The suggested detection algorithm has a cap on system resources. As a result, our solution has a lower time complexity and error rate than existing approaches, being merely (log n) in terms of time complexity. As a result, the method is able to identify denial-of-service and distributed denial-of-service assaults and shield against SYN flooding attacks (Sun et al., 2022).

Mitigating SYN flood attacks, which are a type of denial-of-service (DoS) attack that targets the TCP (Transmission Control Protocol) SYN handshake process. The authors likely propose a mitigation technique that involves using SYN cookies in programmable data planes to protect against SYN flood attacks. The identification of malicious traffic and its prevention using simplistic connection blocking tactics are made more difficult by the abuse of the widely used TCP as an attack vector. The 10 Gbit/s range of traffic can be handled by modern programmable data plane devices without overflowing. The study demonstrates how their performance might be utilised to protect whole networks from SYN flood assaults. As a result, we examine several protection tactics, including SYN authentication and SYN cookies, and we talk about implementation challenges when transferred to various target data planes, including software, network processors, and FPGAs. Provide performance data for each of the three platforms as well as prototype implementations (Scholz et al., 2020).

Three network-based mitigation approaches for TCP SYN authentication were counter measured, installed, and used in modified forms. They all make use of the TCP three-way handshake procedure to establish a security association with a client before sending its SYN data. Against routine assaults using fake IP addresses, these methods are very successful. The adjustments we've made, however, enable us to block even more advanced SYN floods that may elude most traditional defences. Yet, all future SYN segments do not encounter any appreciable extra latency (0.2ms), at the expense of the delayed first connection attempt. In addition to implementation information with improved security changes, this work gives a thorough explanation and analysis of the methodologies. Implementations are to be deployed in its backbone network and Internet exchange point at NIX.CZ and are constructed on top of the hardware-accelerated FPGA-based DDoS prevention solution created by CESNET (Goldschmidt & Kučera, 2021).

Summary

The literature review on SYN flood attacks and their detection in various papers presents a comprehensive overview of the SYN flood and DOS attack. The review may cover various techniques, tools, and strategies used in previous studies to detect and prevent SYN flood attacks in different network environments, such as cloud computing, VPNs, control systems, web servers, and SDN environments. It highlights the significance of SYN flood attacks as a type of denial-of-service (DoS) attack that targets the TCP protocol by overwhelming the target system with a flood of SYN packets, preventing it from responding to legitimate requests. The review emphasizes the need for accurate and efficient detection of SYN flood attacks to ensure the security and availability of network resources. Various techniques used for detecting SYN flood attacks, including network-based techniques such as sniffing, event-triggered control mechanisms, and machine learning algorithms. The review provides various countermeasures used to prevent and decrease the impact of attack on a target machine. It provides insights into the strengths and weaknesses of these techniques, including their performance metrics, limitations, and future research directions. The effectiveness of different detection techniques, including Wireshark, Tcpdump, Snort, and machine learning algorithms, has been evaluated in terms of accuracy, precision, recall, and F1 score.

3. REQUIREMENTS

3.1 Functional Requirements

The system should be able to generate a high volume of SYN packets targeting a specific IP address or port. It should be able to specify the source IP address, source port, destination IP address, and destination port for the SYN packets. Functional requirements to perform DoS attack using SYN flood includes:

- The ability to generate a high volume of SYN packets targeting a specific IP address or port.
- The ability to initiate SYN flood attacks using various TCP flags, such as SYN, ACK, and RST.
- The ability to customize the rate and intensity of SYN flood attacks.
- The ability to log and analyze the results of SYN flood attacks, including the number of SYN packets sent, responses received, and success rate of the attack.
- The ability to stop the SYN flood attack on demand.

3.2 Non- Functional Requirements

The non-functional requirement for the project includes:

- The ability to generate a high volume of SYN packets targeting a specific IP address or port.
- The ability to initiate SYN flood attacks using various TCP flags, such as SYN, ACK, and RST.
- The ability to customize the rate and intensity of SYN flood attacks.
- The ability to log and analyze the results of SYN flood attacks, including the number of SYN packets sent, responses received, and success rate of the attack.
- The ability to stop the SYN flood attack on demand.

3.3 Operational Requirements

A controlled test environment be set up for conducting the SYN flood attack, which include one target system along with any necessary hardware, software resources. The test environment be isolated from production systems to prevent any unintended impact on real systems or networks. Virtual to simulate and test the DoS attack using SYN flood. Setting up a local network or virtual network environment using VMware. An appropriate operating system to run attack simulation, such as Linux or Windows. Linux-based distributions Kali Linux used

for security testing due to their built-in tools and utilities for network analysis and penetration testing. Packet generation tools, network monitoring and analysis tools, be required for conducting the SYN flood attack and analysing network traffic such tools are Nmap, hping3, Wireshark.

Hping3

The attacker sends packets to the victim using Hping3, and these packets are fabricated, crafted, and in a unique form as seen in Figure 1(Islam et al., 2021).

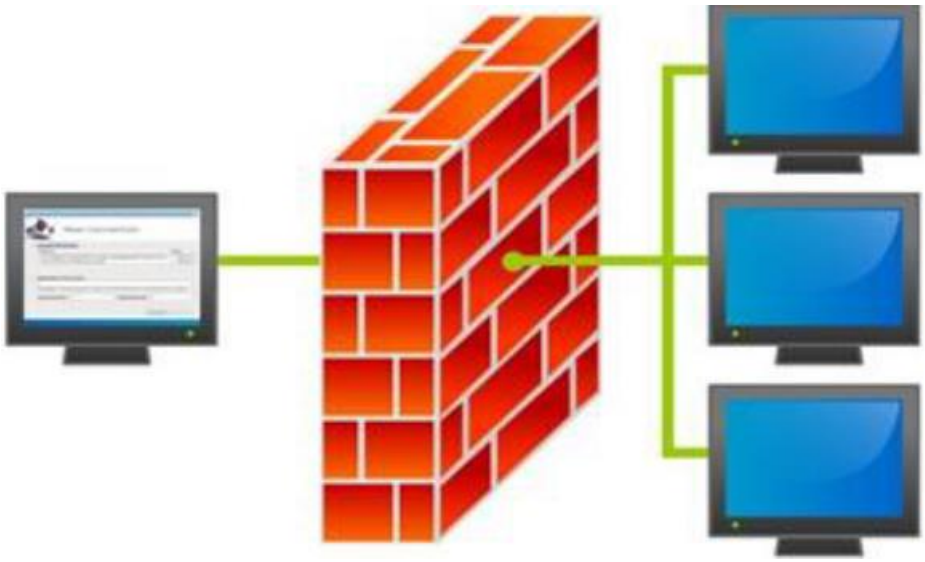

Figure 4: DoS Attack with Hping3 (Islam et al., 2021)

Discovering the MTU path Use is appreciated for fire walk. Contemporary port scanning Net Several protocols, packet sizes, TOS (a category of service), and fragmentation are used to test performance. Using several protocols, traces routes similar. even in the face of fascist firewall regulations, file transfer. verify the firewall rules. Sending a specifically designed payload to web servers or other systems is one way to start the assault (Islam et al., 2021). Hping3 can be used to simulate a SYN flood attack by crafting and sending a large number of TCP SYN packets to a target system. This can help in testing the resilience of a system to SYN flood attacks and evaluating the effectiveness of mitigation measures such as rate limiting, synproxy, or other techniques (sharma, 2020).

Nmap

Nmap, which stands for Network Mapper, is a free and open source programme used for port scanning, vulnerability analysis, and, obviously, network mapping. It is used for network

exploration, security auditing, and vulnerability assessment. It allows users to discover hosts, services, and open ports on a computer network. Nmap can be used to scan for known vulnerabilities in hosts and services, which can help in identifying potential weaknesses that could be exploited in SYN flood attacks. This can help in prioritizing security measures and patching vulnerable systems to prevent SYN flood attacks. With the use of Nmap, hosts and services may be scanned for known vulnerabilities, which can assist find possible flaws that might be leveraged in SYN flood attacks. In order to avoid SYN flood attacks, this can assist in setting security precautions as a priority and updating susceptible systems. Nmap can be used to scan for open ports on a target system, including the destination ports used in SYN flood attacks. This can help in identifying open ports that are susceptible to SYN flood attacks and take appropriate measures to secure them, such as implementing rate limiting, synproxy, or other mitigation techniques (Shivanandhan, 2020).

Wireshark

Wireshark allows capturing and analysing network traffic, including SYN packets, during the SYN flood attack. It provides detailed information about the captured packets, including source and destination IP addresses, port numbers, protocol, and packet contents, which can be used to analyse the behavior of the attack and understand its impact on the targeted system. Wireshark can be used to capture and analyse both incoming and outgoing SYN packets during the SYN flood attack. It can help identify patterns or anomalies in the network traffic, detect any abnormalities in the SYN packet exchange process, and analyse the rate and volume of SYN packets being sent and received shown in Figure 5. This information can be used to assess the effectiveness of the SYN flood attack and make adjustments to the attack parameters, such as the rate of SYN packet generation, to optimize the attack (omerros, 2023).

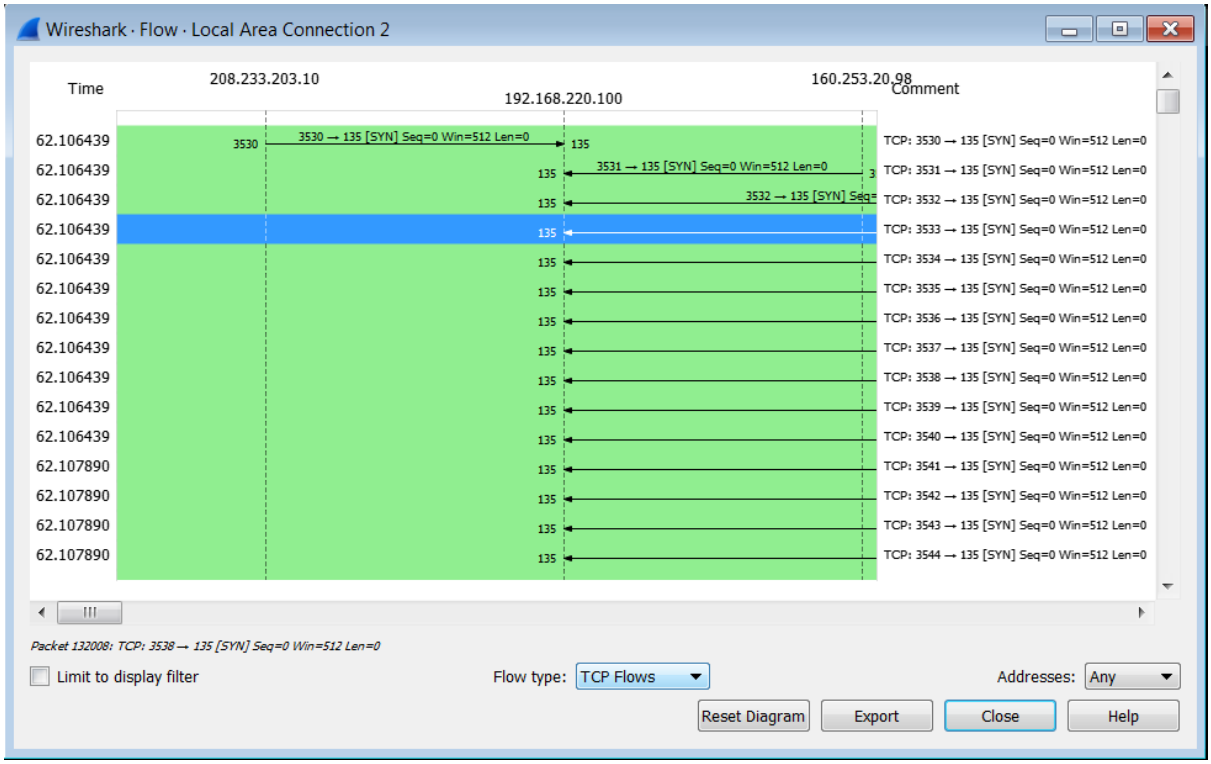

Figure 5: Screenshot of TCP Flow Graph

Wireshark can be used to capture and analyse other network traffic related to the SYN flood attack, such as ICMP messages, TCP ACK or RST packets, and other network events that may occur during the attack. This can provide insights into the behaviour of the targeted system and help in understanding its response to the SYN flood attack (omerros, 2023).

4. METHODOLOGY

Agile methodology can be very effective for this project focused on addressing security vulnerabilities such as the denial-of-service (DoS) attack by SYN flood. The product backlog is a dynamic list of features, requirements, and tasks needed to address the SYN flood attack. It is continuously updated and prioritized based on the criticality of vulnerabilities identified. Agile methodologies often leverage automated testing tools and techniques to streamline the testing process and identify vulnerabilities more efficiently. Automated testing can include techniques such as penetration testing, vulnerability scanning, and security testing tools that simulate SYN flood attacks to identify potential weaknesses in the system. Automated testing enables the project team to quickly identify vulnerabilities, prioritize them based on criticality, and address them in a timely manner.

Flexibility: Agile methodologies allow for more flexibility than traditional Waterfall methodologies, which is particularly useful in the context of DoS attacks. Agile methodologies promote prioritization based on value and risk, allowing the project team to allocate resources effectively to address vulnerabilities associated with DoS attacks (Tuovinen & Frilander, n.d.). Because attack patterns can change rapidly, the Agile methodology allows for regular reassessment and adaptation to new information. The flexibility of Agile allows for rapid response to vulnerabilities, as the team can prioritize and address them in a timely manner. This quick response helps to minimize the window of opportunity for attackers to exploit vulnerabilities and mitigate the impact of DoS attacks.

Iterative and Incremental development: Agile methodologies emphasize incremental development, allowing the project team to break the work down into smaller, more manageable tasks (Wrike, 2022). Break down the tasks into smaller chunks, such as scanning, spoofing, flood, half-open connections. Each of these tasks can be developed incrementally, allowing for continuous improvement and feedback. This can be particularly helpful when addressing security vulnerabilities, as it enables the team to tackle one vulnerability at a time, focusing on the most critical issues first.

Frequent testing: Agile methodologies emphasize frequent testing, enabling the project team to identify and address vulnerabilities as they arise (Rigby et al., 2017). This approach can help ensure that the system is continuously improving its defences against SYN flood attacks. The importance of frequent and continuous testing throughout the project lifecycle. This includes testing at various stages, such as unit testing, integration testing, system testing, and acceptance testing. Continuous testing allows the project team to identify and address vulnerabilities as they arise, helping to ensure that the system's defenses against SYN flood attacks are continuously improving.

Prioritization: Agile methodologies encourage prioritization, which is particularly important when addressing security vulnerabilities. To identify the most critical vulnerabilities and focus on addressing those first. Prioritization based on risk, which involves assessing the likelihood and impact of vulnerabilities being exploited by attackers. Agile methodologies promote continuous assessment of vulnerabilities and re-prioritization as needed. As the project progresses and new information becomes available, the team can reassess vulnerabilities based on the changing threat landscape, emerging attack patterns, and the evolving system requirements. This allows for regular re-prioritization of vulnerabilities to ensure that the most

critical ones are addressed promptly, and the system's defenses against DoS attacks remain up-to-date.

Agile methodologies provide a flexible and adaptive approach for addressing security vulnerabilities, including those associated with DoS attacks. They emphasize prioritization based on value and risk, continuous assessment and re-prioritization, collaboration and input from stakeholders, and regular feedback and monitoring. The iterative and incremental approach of Agile allows for quick adaptation to changing attack patterns, rapid response to vulnerabilities, collaborative teamwork, continuous improvement, and prioritization based on risk. This flexibility enables the project team to effectively address emerging threats, enhance the system's defenses against DoS attacks, and minimize their impact. By prioritizing vulnerabilities based on their potential impact and likelihood of exploitation, and continuously reassessing and adapting based on changing information, Agile methodologies help ensure that the most critical vulnerabilities are addressed promptly, and the system's defenses against DoS attacks are continuously improved. This approach allows for effective allocation of resources and efforts, proactive risk mitigation, and alignment with the overall security goals of the project.

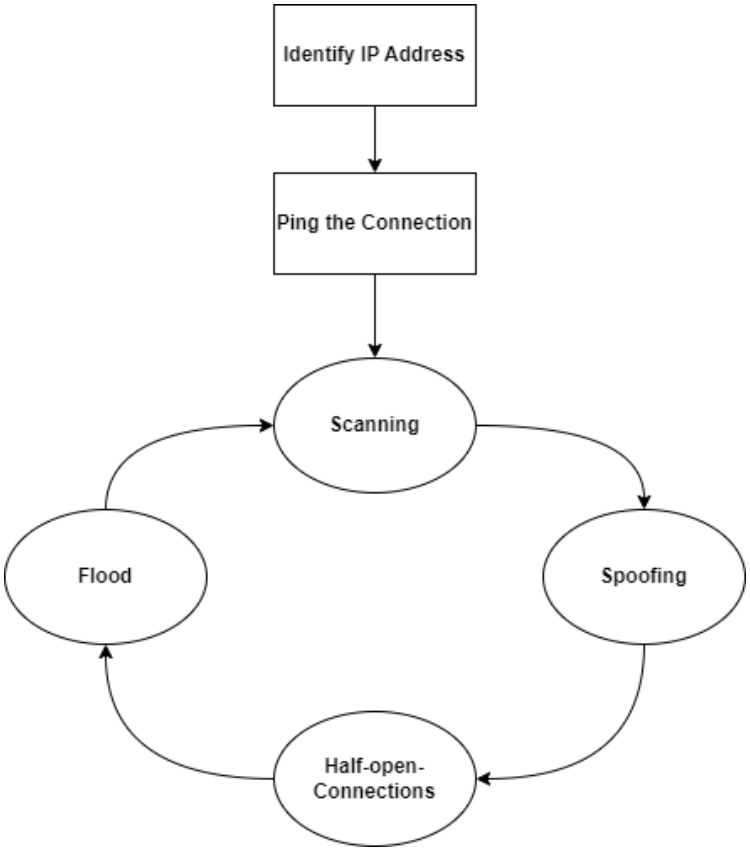

Figure 6: Flow Chart

For conducting a SYN flood attack typically involves the following steps:

Identify IP Address: To perform the denial of service (DoS) attack by SYN flood, it's important to identify the IP addresses of both the target system and the attacking machine. The IP addresses are essential for configuring and launching the SYN flood attack. The IP addresses are used in the configuration of the SYN flood attack, particularly in the spoofing of the source IP address in the SYN packets. Identifying the IP address of both machines to perform the attack, the target system has the IP address 192.168.220.100, and the attacking machine has the IP address 192.168.56.66.

Ping the connection: Check both the machines are communicating with each other or not by using ping command. The ping command can be used as an initial step to check if the attacker's machine and the target system are communicating with each other. The ping command can be used to verify the connectivity between the attacker's machine and the target system and ensure that they are able to exchange ICMP Echo Request and Echo Reply messages. if the target system is online and reachable from the attacker's machine. This can help in determining if the target system is vulnerable to a SYN flood attack, as a successful response to the ping command indicates that the target system is online and can potentially receive and process network traffic, including SYN packets.

Scanning: Perform scanning as one of the initial steps to identify potential vulnerable systems that can be used to launch the attack. Port scanning is a common scanning technique used to identify open ports on a target system. Nmap (Network Mapper) is a widely used tool for performing port scanning. It provides a variety of scanning techniques and options that can be used to identify open ports on target systems. Nmap can be used to scan for open ports on a range of IP addresses, specific hosts, or entire networks. It supports various scanning techniques, such as TCP SYN scan, TCP connect scan, UDP scan, and more. By identifying open ports, it can determine which systems are running services that may be susceptible to a SYN flood attack, such as web servers, mail servers, or other network services. to identify the specific vulnerabilities or weaknesses associated with those services that could be exploited in a SYN flood attack. This information can be used to prioritize and implement appropriate security measures, such as patching vulnerabilities, updating configurations, or implementing network security controls, to protect against SYN flood attacks and other potential security threats. Regular and periodic scanning can also be performed to continuously assess the

security posture of the target systems and identify any changes that may introduce new vulnerabilities or weaknesses.

Spoofing: Then spoofs the source IP address in the SYN packets used in the attack. This involves modifying the source IP address field in the IP header of the SYN packets to make them appear as if they are coming from legitimate sources, such as trusted servers, public IP addresses, or other legitimate sources on the internet. By spoofing the source IP address, the attacker can hide their true location and make it difficult for the target system to trace the attack back to its origin. By spoofing the source IP address in the SYN packets, the attacker can make it appear as if the packets are coming from multiple legitimate sources, distributed across different IP addresses or networks. This can make it challenging for the target system to differentiate between legitimate and malicious traffic, as the attack traffic may appear similar to legitimate connection requests.

Flood: Then generates a large number of SYN packets, which are the initial packets of the TCP handshake process used to establish a connection between a client and a server. These SYN packets typically contain a random source IP address and port number, and they are sent to the target system with the aim of initiating a connection. The SYN packets are sent in rapid succession, often at a high rate, to flood the target system with a large volume of incoming SYN requests. The aim is to overwhelm the system's resources, such as its CPU, memory, and network bandwidth, with an excessive number of half-open connections, where the system has received SYN packets but has not yet completed the handshake process by sending back the corresponding ACK packets.

Half-Open Connections: When the target system receives the spoofed SYN packets, it responds with SYN-ACK packets to the supposed source IP addresses. However, since the attacker has spoofed the source IP addresses, the SYN-ACK packets cannot reach the legitimate sources, as they do not exist or do not respond. As a result, the connections are left in a half-open state, where the target system is waiting for the ACK packets to complete the handshake process, but they never arrive. The half-open connections consume system resources, such as memory, CPU, and network bandwidth, as the target system continues to wait for the ACK packets that will never arrive. With a large number of half-open connections created by the SYN flood attack, the system's resources can quickly become exhausted, resulting in a denial of service (DoS) condition. The system may slow down, become unresponsive, or crash, causing service disruptions or downtime.

After the initial SYN flood attack, repeat the attack multiple times, either sequentially or simultaneously from multiple sources. This is done to sustain the attack over an extended period of time and keep the target system continuously overwhelmed with a high volume of SYN packets. Also vary the attack parameters, such as the rate of SYN packets, the source IP addresses used for spoofing, the target ports, and the duration of the attack, to evade detection and mitigation efforts by the target system's defences.

5. IMPLEMENTATION

Set up a Lab Environment

Selecting the target system as Windows to simulate a real-world target. To protect other systems from any potential threat, this system will be isolated from the rest of the network. Kali Linux be the attacking machine. The operating systems are run on the Virtual machine. Install and configure a packet capture tool Wireshark on the Windows system. This tool helps to capture and analyse the network traffic during the attack. To perform the attack SYN flood tool Hping3 is used which is open source and in-built in Kali Linux. Nmap is used to identify the open ports on the Windows system (target machine). The attacking machine (Kali) uses the hping3 command to produce the SYN packets. A utility called Hping3 can produce more SYN packets in a given amount of time (Wireshark, 2016). When the target machine gets SYN packets, a connection cannot be made to a specific server or system at that time, but the Wireshark tool is still able to record network activity. The Wireshark log file is useful for identifying network activity and behaviour (Bagyalakshmi, n.d.).

Simulation of TCP SYN Flood Attack

Windows and Kali Linux are installed in VMware. Windows machine acted as a target machine and Kali Linux acted as an attacking machine. Normal connection using ping command on both machines. The Ping command is used on both machines to check the network connection. VMware provided virtual adapter connects the two machines.

The first step is to open the cmd prompt in the virtual Windows that has been installed and run ipconfig to find out the IP address of the target machine, as shown in Figure 7.

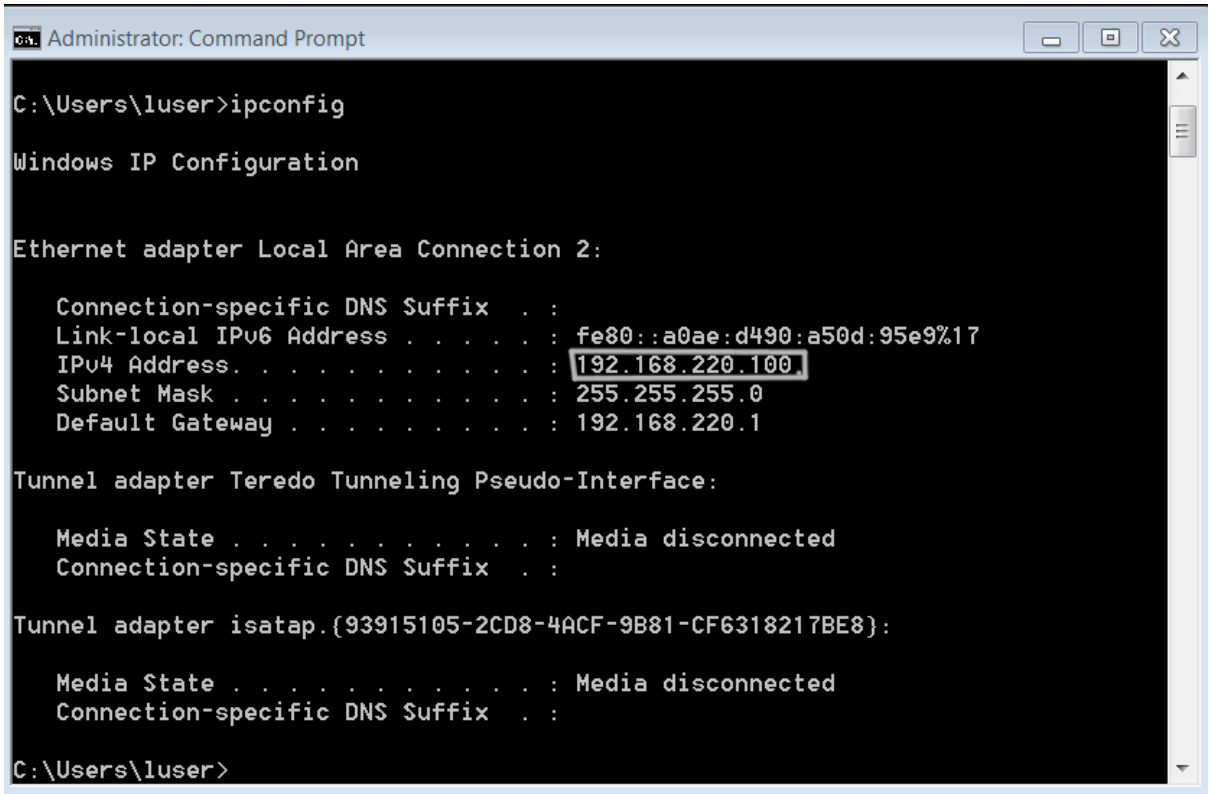

Figure 7: IP address (Victim address)

The next step is to identify the IP address of attacking machine. In order to know the Ip address of kali machine type ifconfig (kali doesn't recognize command ipconfig) as in Figure 8. Now, we have two different IP addresses. one is the address of the Victim machine that is Windows 7 and the other is of the attacker machine i.e. Kali. Once the IP addresses of both the target system (Windows 7) and the attacking machine (Kali Linux) are identified, they can be used in the subsequent steps of the attack process, such as crafting and sending the SYN flood packets with the spoofed IP address, initiating the attack, and monitoring the impact on the target system.

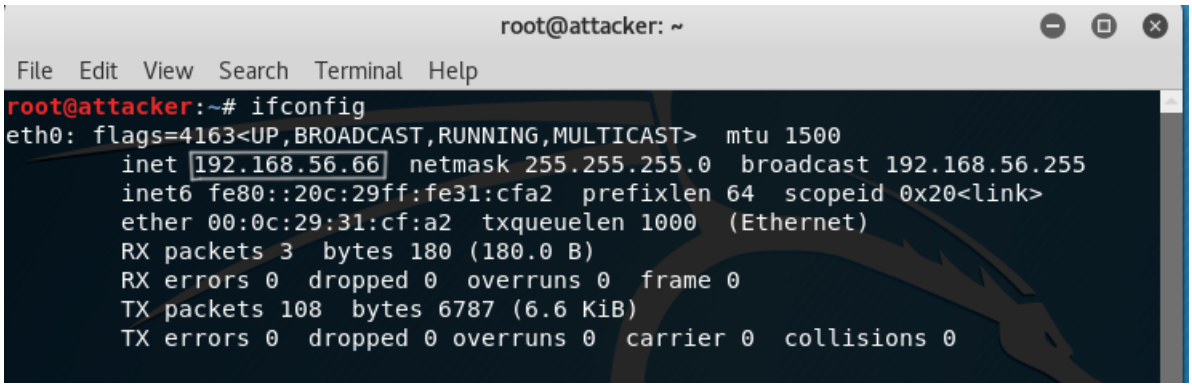

Figure 8: IP address (attacking machine)

After obtaining the IP address of the target system (Windows 7) from the previous steps, the "ping" command is executed in the terminal of the attacking machine (Kali Linux) followed by the IP address of the target system. To verify whether the two computers are in communication with one another or not. The "ping" command is a network utility that sends ICMP Echo Request packets to the specified IP address and waits for ICMP Echo Reply packets in response. This is a common method to test network connectivity between two devices. To find out, type ping followed by the IP address of the other computer (Windows 7) shown in Figure 9 to perform the test. If the communication between the target system and the attacking machine is successfully established, the output of the "ping" command show responses with round-trip time (RTT) and zero percent packet loss. This indicates that the two machines are in communication with each other and can exchange network packets.

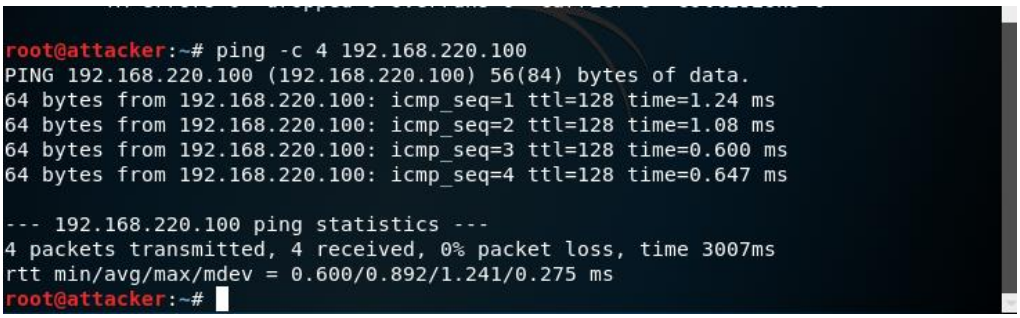

Figure 9: Pinging Target IP Address

The open source and user-friendly tool Wireshark are used to analyse network and packet statistics. Also, it aids in identifying and improving the flow of packets, packet information, protocol information, etc. Identifying the open ports of the victim machine by using the tool Nmap. Nmap is used to identify the open ports shown in Figure 10, it is scanning the target machine (windows) get the information of the ports which can be compromised. The attack is performing on the open port number135.

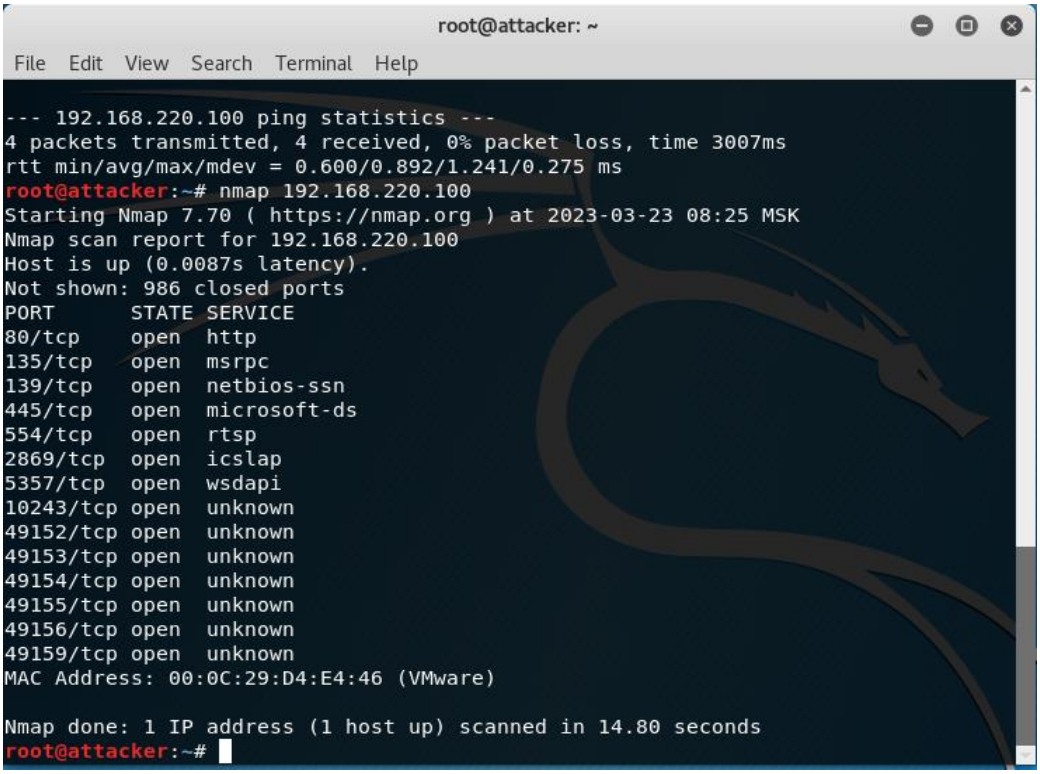

Figure 10: Identifying open ports in the target machine.

Attacking machine is ready to generate and send the SYN packets to the target machine shown in Figure 11. The Hping3 tool, which is available in Kali Linux, is used to generate the SYN packets. Hping3 is a powerful network tool that allows for various types of packet manipulation and generation, including sending SYN packets to initiate a SYN flood attack.

Hping3 command for generating the packet:

hping3 -V -S -p <portno of targetmachine> --flood --rand-source <victim's IP address>

Hping3 command with appropriate parameters to generate a large number of SYN packets and send them to the target system's IP address. The SYN packets are typically generated with spoofed source IP addresses to hide the true identity of the attacker. SYN packets were successfully generated by the Hping3 command and reached the victim system.

Once the Hping3 command is executed, the SYN packets are generated and sent to the target system. These SYN packets overwhelm the target system's resources, causing it to become overwhelmed and unable to respond to legitimate connection requests. As a result, the victim system may experience network downtime or disruption. Unless the Hping3 command is halted, the victim system could not access their network activities.

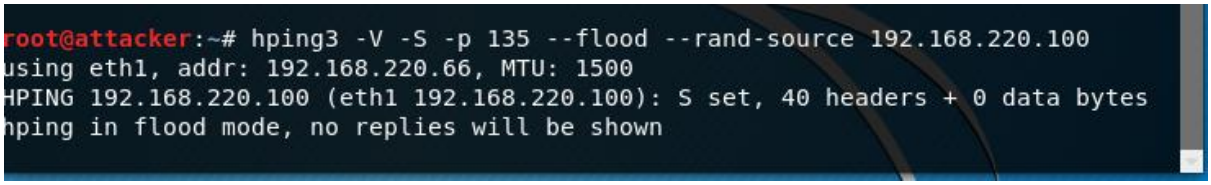

Figure 11: Performing SYN flood attack by Hping3 Tool

To analyse network behaviour, the Wireshark tool generates and captures SYN packets shown in Figure 12. These SYN packets are typically part of the initial three-way handshake process in establishing a TCP connection, where the client sends a SYN packet to the server, the server responds with a SYN-ACK packet, and the client sends an ACK packet to complete the handshake. One of the DDoS attacks is the SYN attacks which is protocol-based, consumes more bandwidth, delays the service, and so on.

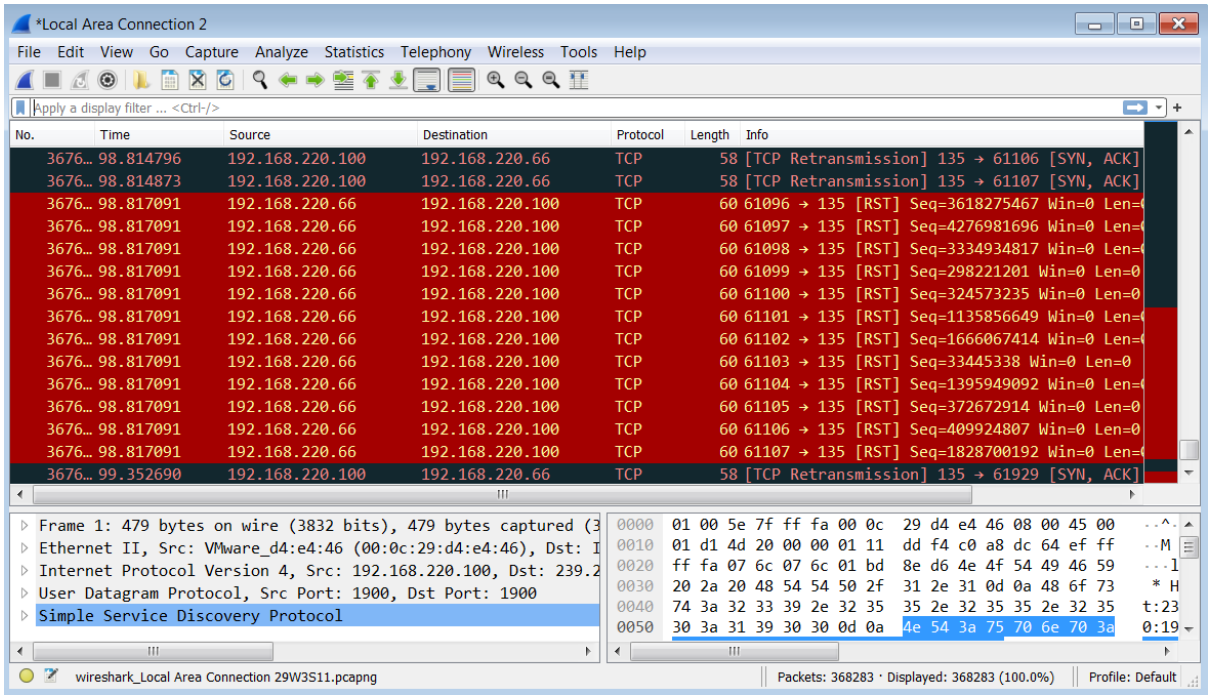

Figure 12: Network Captured by Wireshark (after attack scenario)

6. RESULT AND ANALYSIS

From implementation, there were three main metrics were considered in results. The first factor is the average RTT (Round Trip Time), which measures how long it takes for a packet to travel from its source to its destination before returning. RTT in action via the time difference between a request and a response. For establishing TCP connections, RTT is a crucial measure (Kolahi et al., 2014). The other metric is CPU utilization of the victim, the use of CPU frame length and current rate shows a significant difference between the usual scenario and the attacked

scenario, as illustrated in figures 13 and 14. The bandwidth was the last metric measurement. The network is impacted by the flood attack since it uses up bandwidth, as was previously indicated. then evaluate the defences using these criteria.

RTT, CPU usage, and bandwidth were measured after 5 minutes of traffic generation (one run). The runs were continued for more than 20 times, and the standard deviation was calculated, and the results averaged. Runs kept on until the standard deviation reached 0.05 percent of the average outcomes. Use Wireshark to record the incoming packet and allot the quantity of packets before and after the attack for each run. The attacker generated 128581552 Bytes of traffic while flooding the network with over 25000 packets per second. The result is analysed by implementing two defence strategies for SYN Flood are Rate Liming and SYN Proxy.

Average RTT Results

The average RTT for legitimate user was 1.24 ms before to the attack, but it was about 6245.25 ms during the TCP SYN Attack, as shown in Table 1. The network was jammed by tens of thousands of packets, which caused longer than typical delays.

Scenario	**Average RTT (ms)**
Before Attack	1.24 ms
During Attack	6245.25 ms
With Rate Limiting	3749.45 ms
With SYN Proxy	1524.02 ms

Table 1: Average RTT

The highest outcome as compared to SYN proxy was rate limiting. This occurred because rate limitation merely limits traffic to the threshold; it does not stop malicious traffic. SYN proxy is a more effective defence than the rate limiting. Incoming SYN packets can be intercepted, verified, and only valid SYN packets sent to the destination system. The attacker with a legitimate IP address who may be a zombie was not blocked; only the communication from the fake IP address was.

CPU Utilization

The target system may receive a large number of SYN packets, leading to an overwhelming amount of processing overhead to handle and respond to these packets. The CPU may be utilized heavily to process incoming SYN packets, generate SYN-ACK responses, and manage

the connection state table. This can result in high CPU utilization during the attack, potentially causing the system to slow down, become unresponsive, or even crash due to resource exhaustion.

Scenario	**Utilization**
Before Attack	0%
During Attack	100%
With Rate Limiting	9%
With SYN Proxy	2%

Table 2: CPU Utilization

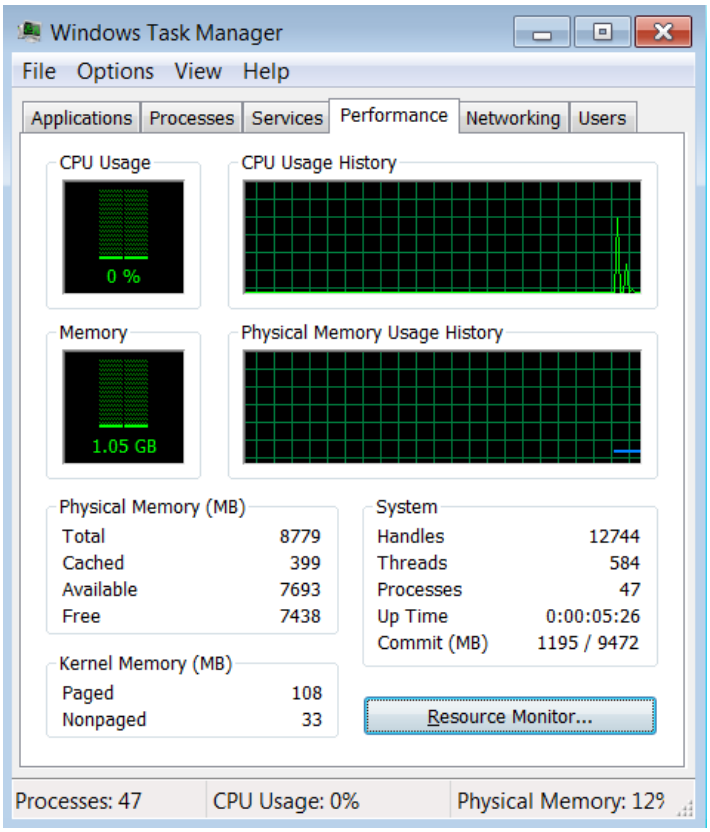

Figure 13: CPU utilisation before attack scenario

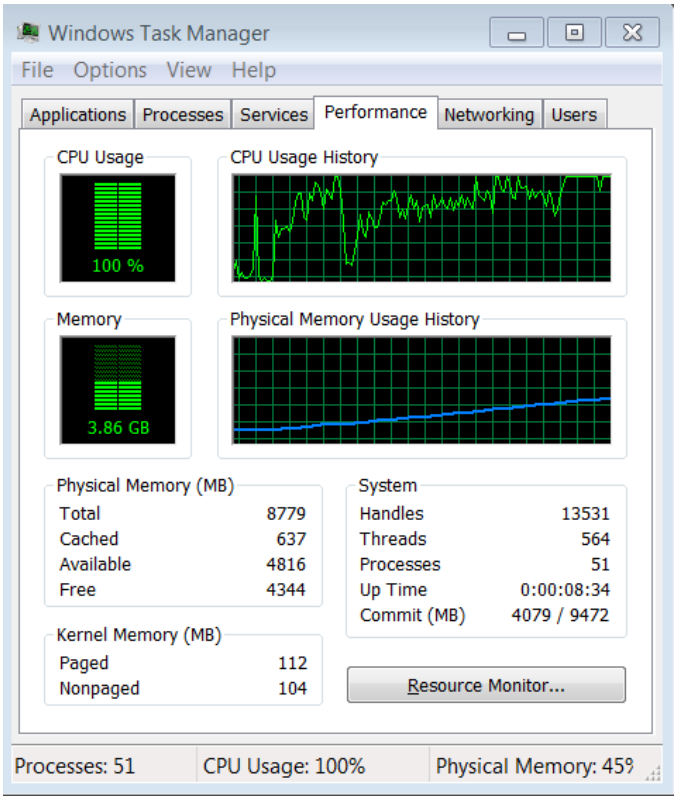

Figure 14: CPU utilisation after attack scenario

Before the attack, there was 0% CPU utilisation shows in figure 13, but during the TCP SYN Attack, that number increased to 100% shown in Table 2. When it came to defence, it was revealed that SYN proxy had totally stopped malicious traffic. It is able to transfer the target system's processing of incoming SYN packets to the SYN proxy. Incoming SYN packets may be verified, SYN-ACK answers can be produced, and the connection status table can be managed by the SYN proxy, which lessens the processing load on the target system. As comparison to a scenario in which there is no defensive mechanism, the CPU utilisation on the target system may be reduced as a result. While rate limiting permits packet passage and some packets do manage to get through, rate limiting has a CPU utilisation of 9%.

Average Network Bandwidth

As the volume of traffic increases, the utilization of bandwidth also increases proportionally. As seen from Table 3, the bandwidth traffic rate without the attack is recorded at 1.255Kbps. However, during the attack, this rate has significantly risen to 845.49Kbps, causing difficulties for legitimate users in accessing the resources. SYN proxy has the highest traffic rates which is 647.01.

Scenario	Traffic Rates (Kbps)
Before Attack	1.255
During Attack	845.49
With Rate Limiting	425.07
With SYN Proxy	647.01

Table 3: Average Traffic Rates

SYN proxy itself may introduce additional overhead due to the processing and inspection of incoming connections. This can result in increased traffic rate between the attacker and the SYN proxy, as well as between the SYN proxy and the target system. Rate Limiting, with a traffic rate bandwidth of 425.07 Kbps, permits traffic within a certain threshold limit, allowing some traffic to pass through while limiting the overall rate of traffic.

Countermeasures

Countermeasures are essential to effectively defend against SYN flood attacks and mitigate their impact.

a. Firewall Rules:-

Implementing firewall rules is an effective strategy to mitigate SYN flood attacks. Firewalls act as a barrier between a trusted internal network and untrusted external networks, filtering incoming and outgoing network traffic based on predefined rules (Mihalos et al., 2019). Steps to implement firewall rules to mitigate SYN flood attacks:

Identify and block malicious IP addresses: Analyse network traffic and identify the IP addresses that are the source of SYN flood attacks. These IP addresses can be added to the firewall's blocklist to prevent further traffic from reaching the target system.

Allow only legitimate traffic: Configure firewall rules to allow only legitimate traffic, such as traffic from trusted sources or specific IP addresses that are required for legitimate communication with the target system. This can be done by setting up whitelist rules in the firewall, allowing only approved sources to establish connections.

Drop excessive SYN packets: SYN flood attacks involve sending a large number of SYN packets without completing the TCP handshake, resulting in half-open connections. Firewall

rules can be configured to detect and drop excessive SYN packets, preventing the target system from being overwhelmed with such traffic.

Enable connection tracking: Many modern firewalls support connection tracking, which allows them to keep track of the state of network connections. Enabling connection tracking can help detect and block SYN flood attacks, as it allows the firewall to track the state of incoming SYN packets and only allow those that are part of a legitimate connection.

Configure rate limiting: Firewall rules can be configured to limit the rate of incoming SYN packets from a single IP address within a specified time frame. This can help prevent the accumulation of half-open connections caused by SYN flood attacks, by blocking excessive SYN packets from a single source.

Regularly update firewall rules: Firewall rules should be regularly updated to adapt to changing threats and to ensure that new malicious IP addresses or patterns of attack are blocked. Regularly review and update firewall rules based on the latest threat intelligence and security best practices.

Test and verify firewall rules: After implementing firewall rules, thoroughly test and verify their effectiveness in mitigating SYN flood attacks. Conduct regular penetration testing and vulnerability assessments to identify any potential gaps in the firewall rules and promptly address them.

Monitor firewall logs: Regularly monitor firewall logs to detect any signs of SYN flood attacks or other malicious activity. Set up alerting mechanisms to notify security personnel of any suspicious network traffic patterns.

Follow the principle of least privilege: Follow the principle of least privilege when configuring firewall rules, only allowing necessary network traffic and blocking all other traffic by default. This can minimize the attack surface and reduce the risk of SYN flood attacks.

Implementing firewall rules is a critical mitigation strategy to defend against SYN flood attacks. By carefully configuring and regularly updating firewall rules can effectively block malicious traffic, limit the impact of SYN flood attacks, and protect target systems from such cyber threats.

b. SYN Cookies:-

Applying SYN cookies is a technique that can be used on the target system to protect against SYN flood attacks. SYN cookies are a form of defence mechanism that involves generating a unique token embedded in the SYN-ACK response to verify the legitimacy of incoming connection requests (Shen et al., 2021). The steps to apply SYN cookies for mitigating SYN flood attacks:

Enable SYN cookies: SYN cookies need to be enabled on the target system. This can typically be done through system configuration settings. SYN cookies are usually implemented at the operating system level, and different operating systems may have different methods of enabling them.

Configure SYN cookie parameters: SYN cookie parameters need to be configured based on the specific requirements of the target system. These parameters may include settings such as the maximum number of allowed half-open connections, the length of time for which SYN cookies are valid, and the format of the SYN cookies.

Generate and include SYN cookies in responses: When the target system receives a SYN packet, it generates a SYN cookie, which is a unique token embedded in the SYN-ACK response. The SYN cookie typically contains information such as the timestamp, source IP address, and a secret key.

Verify SYN cookies in subsequent packets: When the client receives the SYN-ACK response with the SYN cookie, it verifies the legitimacy of the connection request by computing the expected value of the SYN cookie based on the same algorithm used by the target system. If the computed value matches the value in the received SYN cookie, the connection is considered legitimate, and the client sends an ACK packet to complete the TCP handshake.

Discard invalid SYN packets: If the SYN cookie verification fails, indicating that the incoming connection request may be part of a SYN flood attack, the target system can simply discard the invalid SYN packet, preventing the accumulation of half-open connections.

Monitor and adjust SYN cookie settings: Regularly monitor the effectiveness of SYN cookies in mitigating SYN flood attacks and adjust the SYN cookie settings as needed. This may involve fine-tuning parameters such as the maximum number of allowed half-open connections or the length of time for which SYN cookies are valid, based on the traffic patterns and requirements of the target system.

Test and verify SYN cookie implementation: Thoroughly test and verify the implementation of SYN cookies to ensure that they are functioning correctly and effectively mitigating SYN flood attacks. Conduct penetration testing and vulnerability assessments to identify any potential weaknesses or gaps in the SYN cookie implementation and address them promptly.

Applying SYN cookies can be an effective defence mechanism against SYN flood attacks as it helps the target system differentiate between legitimate and malicious connection requests. However, it's important to configure, monitor, and test SYN cookie implementation to ensure its effectiveness and compatibility with the target system.

7. PROJECT MANAGEMENT

A comprehensive literature review to understand various SYN flood attack techniques, including their types, characteristics, and impacts. Evaluate the effectiveness of different SYN flood attack techniques in disrupting the availability and performance of targeted systems or networks. Identify the vulnerabilities and weaknesses in target systems or networks that make them susceptible to SYN flood attacks. Develop a test environment to simulate SYN flood attacks using appropriate tools or software. Conduct controlled experiments to simulate the SYN flood attacks on target systems. Monitor and measure the effects of SYN flood attacks, such as network congestion, service degradation, and system downtime. Research and analyse various mitigation techniques and countermeasures against SYN flood attacks, including firewalls, load balancers, rate limiters, and other security measures. Evaluate the effectiveness of different mitigation techniques in preventing or mitigating SYN flood attacks and compare their strengths and weaknesses. Develop recommendations for implementing effective SYN flood attack mitigation techniques based on the findings and results.

Analysing SYN flood attack techniques and evaluating mitigation techniques in a controlled environment. The project will not involve any real-world SYN flood attacks on actual systems, and all experiments will be conducted in a controlled and ethical manner. The use of simulation tools or software to replicate SYN flood attacks and measure their effects, without causing harm or disruption to any real systems or networks. The project will primarily focus on SYN flood attacks and may not cover other types of DoS attacks or related cybersecurity topics in depth. Aim to provide recommendations and insights based on the findings and results obtained from the simulations.

7.1 Project Schedule

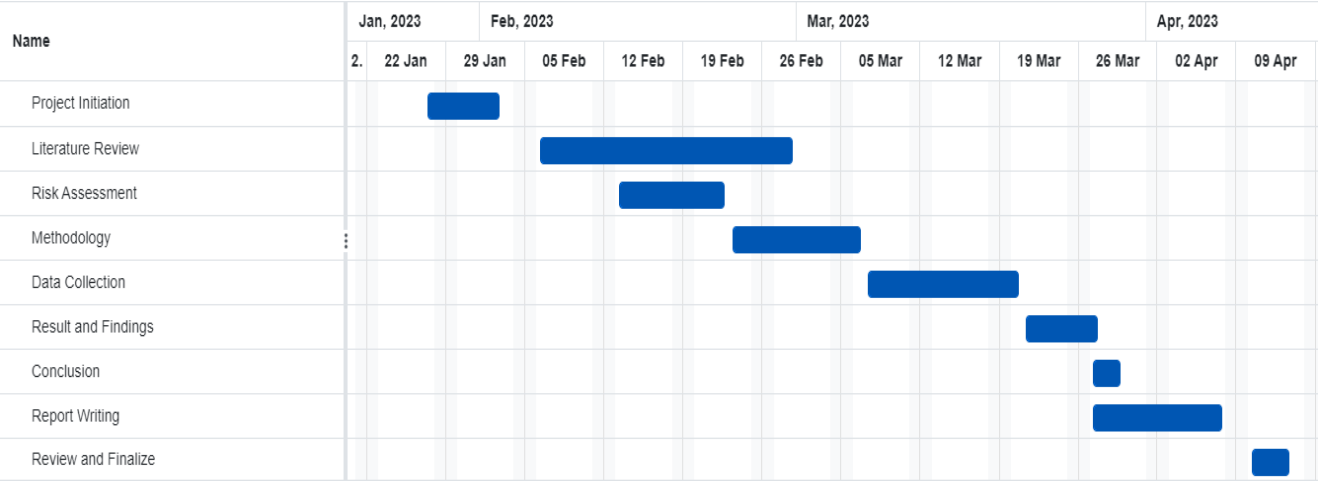

Figure 15: Gantt Chart

Figure shows the project schedule and timeline. It is a graphical representation of project tasks, their durations, and their dependencies.

7.2 Risk Management

Identify potential risks that, including technical, operational, financial, and organizational risks. Risks associated with SYN flood attacks could include network downtime, data loss, reputation damage, and financial losses. Risks involve vulnerabilities in the technical infrastructure and systems that may be exploited by SYN flood attacks.

- Vulnerabilities in network devices, servers, or applications that could be targeted by SYN flood attacks.
- Inadequate or outdated security measures, such as firewalls, intrusion detection systems, or rate limiting mechanisms that may not effectively prevent SYN flood attacks.
- Lack of redundancy and backup mechanisms that could result in data loss or prolonged downtime in the event of a successful SYN flood attack.
- Complexity and scalability issues in the project's technical architecture that may make it susceptible to SYN flood attacks.

Risks involve operational aspects of the project that may impact its ability to detect, prevent, and respond to SYN flood attacks.

- Vulnerabilities in network devices, servers, or applications that could be targeted by SYN flood attacks.
- Inadequate or outdated security measures, such as firewalls, intrusion detection systems, or rate limiting mechanisms that may not effectively prevent SYN flood attacks.
- Lack of redundancy and backup mechanisms that could result in data loss or prolonged downtime in the event of a successful SYN flood attack.
- Complexity and scalability issues in the project's technical architecture that may make it susceptible to SYN flood attacks.

Risks involve potential financial impacts of SYN flood attacks,

- Vulnerabilities in network devices, servers, or applications that could be targeted by SYN flood attacks.
- Inadequate or outdated security measures, such as firewalls, intrusion detection systems, or rate limiting mechanisms that may not effectively prevent SYN flood attacks.
- Lack of redundancy and backup mechanisms that could result in data loss or prolonged downtime in the event of a successful SYN flood attack.
- Complexity and scalability issues in the project's technical architecture that may make it susceptible to SYN flood attacks.

Mitigation Strategy

Maintain the most recent security patches and upgrades on all systems, network devices, and apps to address known vulnerabilities that SYN flood attacks can exploit. Install strong firewalls, intrusion detection and prevention systems (IDPS), rate limiting devices, and other security measures to recognise and stop SYN flood attacks. Reduce the impact of a SYN flood on a single server by distributing incoming SYN requests over several servers using load balancers or reverse proxies. Limit the amount of concurrent incomplete connections (half-open connections) and lower the likelihood of SYN flood assaults by configuring systems and network devices. Two mitigation strategies are implemented SYN Proxy and Rate limiting.

a. SYN Proxy:

Deploy a SYN proxy that can intercept incoming SYN packets and validate them before forwarding them to the target system. This can help filter out illegitimate SYN packets and

protect the target system from the flood of SYN packets. SYN proxy acts as an intermediate layer between the client and server, intercepting incoming SYN packets and validating them before forwarding them to the server. It helps to mitigate SYN flood attacks by ensuring that only legitimate SYN packets are forwarded to the server, while dropping or delaying malicious SYN packets (Dang et al., 2018). Implementing SYN proxy involves following steps:

Deploy synproxy: Install and configure a synproxy tool or software on the targeted system. There are various open-source and commercial synproxy tools available, SYNPROXY which can be used to implement synproxy functionality.

Configure synproxy settings: Configure the synproxy settings based on the specific requirements of the environment. This may involve specifying the allowed rate of incoming SYN packets, setting up SYN cookies, adjusting timeout values, and configuring other parameters that determine the behavior of the synproxy.

Monitor synproxy performance: Monitor the performance of the synproxy in real-time to ensure that it is effectively mitigating SYN flood attacks without impacting legitimate traffic. Use monitoring tools such as traffic analysis, or performance metrics, to assess the effectiveness of the synproxy in mitigating SYN flood attacks.

Fine-tune synproxy settings: Fine-tune the synproxy settings as needed based on observed traffic patterns, system performance, and attack scenarios. Adjust the allowed rate of incoming SYN packets, timeout values, or other parameters as necessary to optimize the performance of the synproxy and improve its effectiveness in mitigating SYN flood attacks.

b. Rate Limiting:

Implement rate limiting mechanisms that restrict the number of connections or SYN packets per second from a single IP address or subnet. This can help prevent the rapid accumulation of half-open connections that can lead to resource exhaustion. Determine the acceptable rate of incoming traffic based on the capacity and capability of the system. This can be done by analysing historical traffic patterns, monitoring system performance, and considering the resources available for processing incoming traffic (El Kamel et al., 2022). Whether to implement rate limiting at the network level, application level, or by using dedicated tools or devices. It should be combined with other security measures, such as firewalls, intrusion detection and prevention systems, and regular security updates, to provide a robust defence against attacks.

7.3 Social, Legal, Ethical and Professional Considerations

Social Considerations:

SYN flood attacks can disrupt the availability and performance of systems or networks, potentially causing inconvenience or financial loss to users, organizations, and other stakeholders. Consider the potential social impact of the project findings and results and take appropriate measures to minimize any negative consequences. Raise awareness about SYN flood attacks and their implications Provide appropriate education and guidance on how to mitigate and respond to such attacks in a responsible and socially conscious manner. This can include educating users about best practices for securing their systems and networks, promoting responsible use of technology, and fostering a culture of cybersecurity awareness. Consider the ethical implications of conducting research or experiments related to SYN flood attacks. Ensure that the project is aligned with social responsibility principles, such as respecting human rights, protecting privacy and confidentiality, and avoiding actions that may cause harm to individuals, organizations, or the society at large. Findings and results can contribute to the development of effective mitigation techniques or best practices to protect against SYN flood attacks, which can benefit not only the specific target systems or networks in the project, but also other systems and networks that may face similar threats in the future.

Legal Considerations:

All applicable rules and regulations, including those governing cybersecurity, data protection, and other pertinent legal frameworks, are observed over the course of the project. Ensures that the methods are in compliance with all applicable local, regional, and federal rules and regulations, including those governing data protection, cybersecurity, and computer crimes. In this work, respect intellectual property rights of others, including copyrights, trademarks, and patents, when using tools, software, or data in the project. Properly attribute and cite any external sources or references used in the project to avoid plagiarism and intellectual property infringements. Obtained all necessary approvals, and permits required by l institutional policies before conducting any activities related to SYN flood attacks. Project activities do not involve any illegal or unethical activities, such as unauthorized access to systems or networks, data breaches, or damage to systems or networks. The legal boundaries and limitations of cybersecurity research and experimentation and ensure that the project activities are conducted within these legal boundaries.

Ethical Considerations:

The project activities respect the rights and dignity of individuals involved, including users, organizations. This includes respecting privacy, confidentiality, and data protection rights, and obtaining informed consent. Project activities are conducted in a fair and equitable manner, without discrimination or bias. Do not manipulating or misrepresent any data or information to achieve desired outcomes in my work.

Professional Considerations:

Maintaining professional conduct throughout in my project, including respectful communication to project supervisor. Continuously update and enhance professional knowledge and skills related to cybersecurity, network security, and project management to ensure high-quality and up-to-date project outcomes. Seek guidance and support from project supervisor as needed.

8. CONCLUSION

An in-depth explanation of the approach and strategies employed in carrying out such attacks is provided by the work on the denial-of-service attack using SYN flood. It covers the frequently used SYN flood attack techniques of scanning, spoofing, flooding, half-open connections, amplification, and repetition. A comprehensive understanding of the nature of SYN flood attacks, their impact on network performance, and various techniques for detecting and preventing them. Highlights the potential impact of SYN flood attacks, which can disrupt the normal operation of a target system or network, causing downtime, loss of productivity, and financial losses. Demonstrate how to set the environment to simulate SYN flood attack, launch using Hping3 tool and analyse the impact of the attack by using Wireshark.

It emphasizes the importance of implementing effective security measures to protect against SYN flood attacks, such as using firewalls, SYN proxy systems, rate limiting, and other security controls. Implementing SYN proxy systems may require additional hardware, software, or configuration changes. It may also introduce additional latency in the connection establishment process, as the proxy system needs to validate incoming SYN packets before forwarding them to the target system. Additionally, SYN proxy systems may require regular updates and maintenance to ensure their effectiveness. Implementing rate limiting may require careful tuning to avoid false positives or false negatives. Too aggressive rate limiting may block legitimate connection requests, resulting in user dissatisfaction or service disruption. On the

other hand, too lenient rate limiting may not effectively mitigate SYN flood attacks, allowing them to overwhelm the target system.

The results shows that impact of attack on the target system and analyse three constraints cpu utilization, network bandwidth and RTT. Based on the results, SYN proxy may be a better mitigation strategy compared to rate limiting, as it can provide more effective offloading of CPU utilization and network bandwidth from the target system, and better filtering of malicious SYN packets. However, the effectiveness of these mitigation strategies may vary depending on the specific network environment, attack patterns, and other factors. Regular monitoring and fine-tuning of mitigation strategies may be necessary to ensure optimal performance and effectiveness in mitigating SYN flood attacks. Also analyse the constraints while implementing two mitigation strategies SYN proxy and rate limiting. The results depict that SYN proxy is better mitigation strategy than compared with rate limiting.

8.1 FUTURE WORKS

As attackers constantly evolve their techniques, it is essential to continually develop advanced mitigation techniques to effectively detect and prevent SYN flood attacks. Implementing more advanced defence mechanisms, such as machine learning-based approaches, anomaly detection techniques, or proactive measures to identify and block SYN flood attacks in real-time. Attackers may develop new variants of SYN flood attacks or combine multiple attack vectors to launch more sophisticated attacks. Explore new variants of SYN flood attacks, analysing their characteristics, and devising effective countermeasures to mitigate them. SYN flood attacks may behave differently in different network environments, such as cloud-based environments, Internet of Things (IoT) networks, or industrial control systems (ICS). Evaluate the impact and effectiveness of SYN flood attacks in these unique network environments and developing tailored mitigation strategies.

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APPENDIX A – Meeting Log

Meeting Date	Meeting Time	Agenda
01/02/2023	10:00 – 10:15 AM	Discussed and finalized project topic, Project proposal
15/02/2023	10:00 – 10:15 AM	Discussed about project introduction and literature review.
01/03/2023	10:00 – 10:15 AM	Discussed the feedback of the reviewed literature review. Mentioned to submit methodology.
16/03/2023	10:00 – 10:15 AM	Discuss the feedback of the methodology. Suggested that to include flow chat shows how agile methodology works on.
29/03/2023	10:00 – 10:15 AM	Discuss the methodology and implementation part.
06/04/2023	10:00 – 10:30 AM	Project Presentation via Teams

APPENDIX B – Certificate of Ethics Approval

Performing Denial of Service Attack by SYN Flood P149214

Coventry University

Certificate of Ethical Approval

Applicant:	Keerthi Krishnan
Project Title:	Performing Denial of Service Attack by SYN Flood

This is to certify that the above named applicant has completed the Coventry University Ethical Approval process and their project has been confirmed and approved as Low Risk

Date of approval:	13 Feb 2023
Project Reference Number:	P149214

APPENDIX C – Project Presentation

PERFORMING DOS ATTACK BY SYN FLOOD

7030CEM- Cyber Security Individual Project

Presenting by,

Keerthi Krishnan

Student ID:12639744

Supervisor: Xingang Wang

Slide 1: Title Slide

CONTENTS

- Introduction
- Aim and Objectives
- Literature Review
- Methodology
- Results
- Countermeasures
- Conclusion
- References

Slide 2: Contents

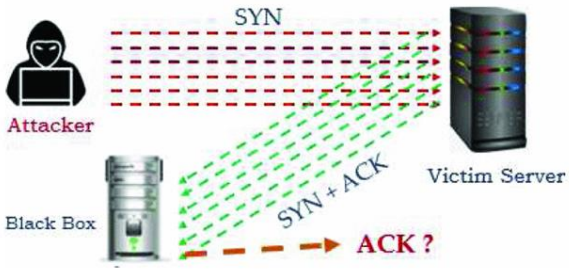
INTRODUCTION

- Denial-of-service (DoS) cyberattack forbids authorised users from using a computer or network. (FRANKENFIELD, n.d.)
- DoS attack typically involves flooding the server or host (memory, CPU, traffic). (Sumantra & Indira Gandhi, 2020)
- SYN flood attacks are a sort of DoS attack that work by flooding a network or system with SYN packets to disrupt or disable it.
- SYN flood attacks can cause network outages, slow down network performance, and even crash servers, which can result in significant financial losses and damage to a company's reputation.

Slide 3: Introduction

SYN Flood

- A large number of SYN packets with spoof IP addresses are sent by the attacker to the victim server.
- A SYN/ACK packet is then returned by the victim server.
- The final ACK from the attacker never replays to the victim server; as a result, a lot of half-open connections are kept in the victim server's queue, and it fills up quickly.



Slide 4: SYN Flood

AIM

- Analyse the DoS attack, with a focus on SYN flood attacks, and their impact on network performance.
- Understanding of SYN flood attacks, their potential risks and threats, and the countermeasures that can be used to prevent or mitigate them.
- A review of the DoS attack and its application using the hping3 tool. This makes it crystal clear how a DoS assault is carried out by an attacker.

Slide 5: Aim

OBJECTIVES

- To understand the basics of SYN flood attacks, including their characteristics, types, and common vulnerabilities in the TCP/IP protocol.
- To set up a lab environment that simulates a network infrastructure and enables the launch of SYN flood attack using hping3 tool.
- The variation is described in the results when the attacks are posed against the restrictions of CPU usage.

Slide 6: Objectives

LITERATURE REVIEW

Using SYN flooding and Metasploit, an overview of DoS attacks has been seen. The effects of a DoS attacks have been examined in terms of CPU execution overhead (Kukreti et al., 2022).

The purposes of comparing the results of the attacks, various commands and approaches have been used. Both Metsploit and SYN flooding are carried out. Sending packets in a modified, crafted form requires the use of Hping3 (Islam et al., 2021).

Slide 7: Literature Review

METHODOLOGY

- Scanning- Scans the target network to identify vulnerable systems that can be used as launching points for the attack.
- Spoofing- Spoofs the source IP address in the SYN packets to make them appear as if they are coming from legitimate sources.
- Flood- Sends a large number of SYN packets to the target system, far more than it can handle.
- Half-open connections- Spoofing the source IP address, the target system will not be able to complete the TCP handshake
- Repeat- Repeat these steps multiple times, with the goal of keeping the target system offline for an extended period.
- Agile approaches offer greater flexibility than conventional Waterfall methodologies, which is very helpful when dealing with DoS attacks.

Slide 8: Methodology

Simulating SYN Flood Attack

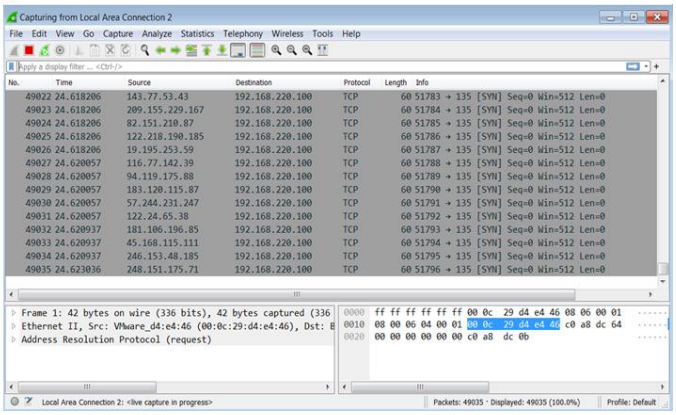
- Windows machine will be the overwhelmed with SYN flood attack using Hping3.
- To protect other systems from any potential threat, the system will be isolated from the rest of the network.
- Identify the open ports on the target machine by using Nmap.
- Wireshark is used to analyse network and packet statistics.

```
root@attacker:~# hping3 -V -S -p 135 --flood --rand-source 192.168.220.100
using eth1, addr: 192.168.220.66, MTU: 1500
HPING 192.168.220.100 (eth1 192.168.220.100): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
```

Slide 9: Simulating SYN Flood Attack

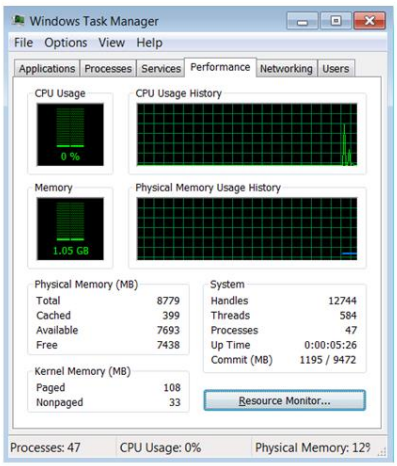
RESULTS

- The SYN flood attack had a significant impact on the target system.
- Response times increased significantly, and error rates also increased, with some requests timing out.

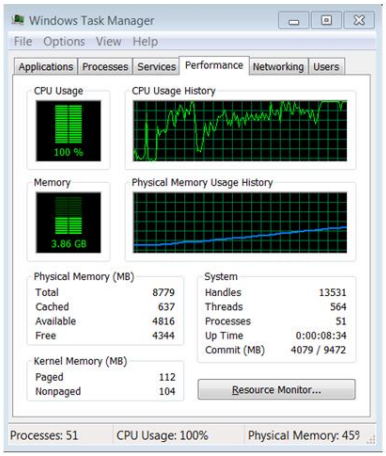


Slide 10: Results

CPU Utilization



Normal Scenario



Attack Scenario

Slide 11: CPU Utilization

COUNTERMEASURES

- To use firewalls to block traffic from known malicious sources.
- Use load balancers to distribute traffic across multiple servers.
- Configuring the system to limit the number of concurrent connections from a single IP address can also help reduce the impact of SYN flood attacks.
- SYN cookies can be used to protect against SYN flood attacks by assigning a unique cookie to each connection request

Slide 12: Countermeasures

CONCLUSION

- Demonstrated the significant impact of a SYN flood attack on a target system, including increased response times, error rates, and network congestion.
- Mitigating the impact of SYN flood attacks requires employing multiple countermeasures, such as firewalls, load balancers, and limiting concurrent connections.
- The impact of a DoS attack on CPU execution overhead has been examined.
- Provides important insights into the risks associated with SYN flood attacks.

Slide 13: Conclusion

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Slide 14: References